

won, but the smartphones and tablets that launched it have remained pretty much the same throughout. In 2013, we saw the emergence of flexible display screens as a viable option for personal electronics. And once the technology is perfected, the range of possibilities gets a whole lot broader. How about a T-shirt that plays YouTube videos? Or a 229cm (90-inch) high-definition TV which you can fold and bring to a friend's house for the big game? All are starting to seem like real possibilities.

The breakthrough came as researchers like those at Colaneri's lab figured out how to build display panels onto paper-thin plastic substrates—thin slices of material that act as semiconductors—instead of glass. "We've always said that flexible displays are sort of the beginning toward truly flexible electronics," he said. "The display in a conventional phone or laptop or tablet is the least flexible thing in there, since it's made out of a piece of rigid glass."

The first wave of consumer gadgets using that and similar technologies is under way, with more promised in the near future.

Korean gadget giants Samsung and LG have already rolled out TVs and phones with curved display screens. Apple has filed for a patent for an 'electronic device with a wraparound display.' At Google, one of CNN's Thinkers, Mary Lou Jepsen, is a pioneer in the field of display screens. She's in the super-secret Google X lab, so there's no telling what she and others are actually up to. But it's hard to imagine new displays aren't somewhere on the agenda.



Fig. 17: Philips 'fluid' smartphone concept

Trend of flexible electronics market in future

The global flexible electronics market is expected to reach US\$ 25.9 billion by 2018, according to a new report by Global Industry Analysts, Inc., propelled by the increasing demand for lighter and smaller electronic products with low power consumption.

The market, considered to be in an embryonic stage, is currently the focus of intense research by universities, research institutes and technology-driven enterprises. In future, the market will offer immense potential for rapid growth with applications extending beyond the traditional sphere to military, automotive, aerospace, medical and consumer applications.

Displays	Black & white, flexible e-readers	Electrophoretic colour readers	Rollable colour displays
Photo-voltaics	Consumer electronics	Off-grid BIPVs	Grid-connected BIPVs
Batteries	Low-capacity, flexible, discontinuous use	High-capacity, flexible, continuous use	Powering standalone flexible display devices and smart textiles
Sensors	Flexible temperature, pressure sensor arrays	Flexible potentiometric sensor for atmospheric monitoring	Smart sensors
Memory	Gaming, smart-cards, small non-volatile RAM	Medium non-volatile RAM	Large non-volatile RAMs powering the standalone display devices
RFIDs	Brand protection ticketing	Asset tracking, security automation	Item-level tagging
2008	2008	2011 2013 2015	2018

"All of the names that you mentioned are certainly sniffing around (the idea)," Colaneri said. "I think all of the names that you've dropped are also among the top-ten list of most paranoid, secretive organisations known to man. They're certainly thinking about it."

Conclusion

Flexible electronics is an emerging field of science and manufacturing technology, which enables planting of electronic devices onto conformable plastic substrates. The flexible electronics sector, similar to large-area or macro electronics, organic electronics,

plastic electronics and printed electronics verticals, is driven by the global demand for lighter and smaller electronic products that consume lesser power. Due to the fact that these devices are more shock-resistant, cost-effective to manufacture and can be flexed or bended, they have the capability of being integrated into portable devices, clothing and packaging materials.

While the market for flexible electronics is still in its embryonic stage, various new applications are being studied by universities, research institutes and high-tech companies. The technology is finding increasing application in various sectors, such as military, computers, consumer products, aerospace, automotive and medical. The main end-use areas for flexible electronics include electronic displays, photovoltaics, sensors, lighting and logic/memory applications among others. The capability to build flexible electronics, or electronic circuits that are capable of being stretched and bent according to requirement, enables the user to expand the current scope of functions supported by